

Evidence Summary - Technology #12

Osmotic Pressure Regulating Drainage Valve

External Evidence Sources (5 total):

Source 1: Cerebral Spinal Fluid (CSF) Shunt Systems - FDA

URL: <https://www.fda.gov/medical-devices/implants-and-prosthetics/cerebral-spinal-fluid-csf-shunt-systems>

Excerpt: Jul 16, 2018 · Fixed shunt valves allow CSF fluid to drain when CSF pressure exceeds a certain “fixed” threshold. Adjustable shunt valves allow for changes to ...

Source 2: Understanding Shunt Systems | Hydrocephalus Association

URL: <https://www.hydroassoc.org/shunt-systems>

Excerpt: A shunt is a flexible tube placed into the area of the brain called the ventricles, where cerebrospinal fluid (CSF) is accumulating.

Source 3: CSF-Flow Control Valves - Shunt for Hydrocephalus - Medtronic

URL: <https://www.medtronic.com/en-us/l/patients/treatments-therapies/hydrocephalus-shunt/csf-flow-control-valves.html>

Excerpt: A shunt redirects excess cerebrospinal fluid (CSF) in the brain to another part of the body, allowing the brain's enlarged ventricles to return to a more normal ...

Source 4: A Review of Cerebral Shunts, Current Technologies, and Future Endeavors

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6153620>

Excerpt: Sep 21, 2018 · The valve is responsible for the release of CSF from the brain into the second catheter that drains the CSF to the absorption site.

Source 5: Brain shunt - Mayo Clinic

URL: <https://www.mayoclinic.org/tests-procedures/brain-shunt/about/pac-20588547>

Excerpt: Sep 5, 2025 · The valve controls the rate of CSF flow from the brain to the drainage site. This allows the fluid to be absorbed elsewhere and relieves ...

*Evidence Classification: This dossier contains source-native information, derived engineering analysis, engineering proposals, and unresolved questions.
No evidence has been physically validated.*